

Probability & Random Processes EE2011

Department of Electrical Engineering

**Binomial Distribution , Multinomial
Distribution and Hypergeometric Distribution**

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



5.1 Introduction and Motivation

- Binomial distribution is a way to calculate probabilities when there are only two possible outcomes for each event.
E.G.
 - flipping a coin (heads or tails) or
 - test (pass or fail)
- Used when you're looking at the likelihood of a certain number of "successes" (like getting heads) in a fixed number of trials.
- It's like figuring out the probability of getting exactly 3 heads when you flip a coin 5 times.
- **Trials (n):** The number of times the experiment is repeated (e.g., 5 coin flips).
- **Probability of success (p):** The probability of a single "success" (e.g., getting heads) in one trial.
- **Number of successes (k):** The number of successes you want to find the probability for (e.g., 3 heads)



5.1 Introduction and Motivation

No matter whether a discrete probability distribution is represented graphically by a histogram, in tabular form, or by means of a formula, the behavior of a random variable is described.

Often, the observations generated by different statistical experiments have the same general type of behavior.

Consequently, discrete random variables associated with these experiments can be described by essentially the same probability distribution and therefore can be represented by a single formula.

In fact, one needs only a handful of important probability distributions to describe many of the discrete random variables encountered in practice.

Such a handful of distributions describe several real-life random phenomena.

For instance, in a study involving testing the effectiveness of a new drug, the number of cured patients among all the patients who use the drug approximately follows a binomial distribution (Section 5.2).



In an industrial example, when a sample of items selected from a batch of production is tested, the number of defective items in the sample usually can be modeled as a hypergeometric random variable (Section 5.3).

In a statistical quality control problem, the experimenter will signal a shift of the process mean when observational data exceed certain limits.

The number of samples required to produce a false alarm follows a geometric distribution which is a special case of the negative binomial distribution (Section 5.4).

On the other hand, the number of white cells from a fixed amount of an individual's blood sample is usually random and may be described by a Poisson distribution (Section 5.5).

In this chapter, we present these commonly used distributions with various examples.



5.2 Binomial and Multinomial Distributions

An experiment often consists of repeated trials, each with two possible outcomes that may be labeled **success** or **failure**.

We may choose to define either outcome as a success.

The process is referred to as a **Bernoulli process**.

Each trial is called a **Bernoulli trial**.

Strictly speaking, the Bernoulli process must possess the following properties:

1. The experiment consists of repeated trials.
2. Each trial results in an outcome that may be classified as a success or a failure.
3. The probability of success, denoted by p , remains constant from trial to trial.
4. The repeated trials are independent.



Consider the set of Bernoulli trials where three items are selected at random from a manufacturing process, inspected, and classified as defective or nondefective.

The number of defective is a random variable X assuming integral values from 0 through 3.

The eight possible outcomes and the corresponding values of X are

Outcome	<i>NNN</i>	<i>NDN</i>	<i>NND</i>	<i>DNN</i>	<i>NDD</i>	<i>DND</i>	<i>DDN</i>	<i>DDD</i>
x	0	1	1	1	2	2	2	3

Since the items are selected independently and we assume that the process produces 25% defectives, we have

$$P(NDN) = P(N)P(D)P(N) = \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{64}.$$

Similar calculations yield the probabilities for the other possible outcomes.



$$\longrightarrow P(NNN) = P(N)P(N)P(N) = \left(\frac{3}{4}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(NND) = P(D)P(N)P(N) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(NDN) = P(N)P(D)P(N) = \left(\frac{3}{4}\right)\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(NDD) = P(N)P(D)P(D) = \left(\frac{3}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(DNN) = P(D)P(N)P(N) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(DND) = P(D)P(N)P(D) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(DDN) = P(D)P(D)P(N) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$\longrightarrow P(DDD) = P(D)P(D)P(D) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$



$$P(DNN) = P(D)P(N)P(N) = \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{64}.$$

$$P(NDN) = P(N)P(D)P(N) = \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{64}.$$

$$P(NND) = P(D)P(N)P(N) = \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) = \frac{9}{64}.$$

$$P(X=1) = f(1) = P(DNN) + P(NDN) + P(NND) = 3 \times \frac{9}{64}.$$

		<i>DNN</i>	<i>NDN</i>	<i>NND</i>	
<i>x</i>		0	1	2	3
<i>f(x)</i>		$\frac{27}{64}$	$\frac{27}{64}$		



$$P(NDD) = P(N)P(D)P(D) = \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) =$$

$$P(DND) = P(D)P(N)P(D) = \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) =$$

$$P(DDN) = P(D)P(D)P(N) = \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) =$$

$$P(X=2) = f(2) = P(NDD) + P(DND) + P(DDN) = \frac{9}{64} \quad 3 \times \frac{3}{64}$$

			NDD	NDD	DDN	
x	0	1	2	3		
$f(x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$			



The probability distribution of X is therefore

x	0	1	2	3
$f(x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n. \quad p = \frac{1}{4}, \quad q = \frac{3}{4}$$

Thus, for the probability distribution of X , the number of defectives is

$$P(X = 1) = f(1) = b\left(1; 3, \frac{1}{4}\right) = \frac{27}{64}$$



Let us now generalize the above illustration to yield a formula for $b(x; n, p)$.

That is, we wish to find a formula that gives the probability of x successes in n trials for a binomial experiment.

First, consider the probability of x successes and $n - x$ failures in a specified order.

Since the trials are independent, we can multiply all the probabilities corresponding to the different outcomes.

Each success occurs with probability p and each failure with probability $q = 1 - p$.

Therefore, the probability for the specified order is $p^x q^{n-x}$.

This number is equal to the number of partitions of n outcomes into two groups with x in one group and $n - x$ in the other and is written $\binom{n}{x}$.

Because these partitions are mutually exclusive, we add the probabilities of all the different partitions to obtain the general formula, or simply multiply $p^x q^{n-x}$ by $\binom{n}{x}$.



Binomial Distribution

A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Note that when $n = 3$ and $p = 1/4$, the probability distribution of X , the number of defectives, may be written as

$$b\left(x; 3, \frac{1}{4}\right) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3,$$

rather than in the tabular form on page 144.

x	0	1	2	3
$f(x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$



Example 5.1: The probability that a certain kind of component will survive a shock test is $3/4$.

Find the probability that exactly 2 of the next 4 components tested survive.

Solution: Assuming that the tests are independent and $p = 3/4$ for each of the 4 tests, we obtain

$$b\left(2; 4, \frac{3}{4}\right) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2! 2!}\right) \left(\frac{3^2}{4^4}\right) = \frac{27}{128}.$$



Where Does the Name *Binomial* Come From?

The binomial distribution derives its name from the fact that the $n + 1$ terms in the binomial expansion of $(q + p)^n$ correspond to the various values of $b(x; n, p)$ for $x = 0, 1, 2, \dots, n$.

That is,

$$\begin{aligned}(q + p)^n &= \binom{n}{0} q^n + \binom{n}{1} p q^{n-1} + \binom{n}{2} p^2 q^{n-2} + \cdots + \binom{n}{n} p^n \\ &= b(0; n, p) + b(1; n, p) + b(2; n, p) + \cdots + b(n; n, p).\end{aligned}$$

Since $p + q = 1$, we see that

$$\sum_{x=0}^n b(x; n, p) = 1,$$

a condition that must hold for any probability distribution.



Frequently, we are interested in problems where it is necessary to find $P(X < r)$ or $P(a \leq X \leq b)$.

Binomial sums

$$B(r; n, p) = \sum_{x=0}^r b(x; n, p)$$

are given in Table A.1 of the Appendix for $n = 1, 2, \dots, 20$ for selected values of p from 0.1 to 0.9.

We illustrate the use of Table A.1 with the following example.



Example 5.2: The probability that a patient recovers from a rare blood disease is 0.4.

If 15 people are known to have contracted this disease,
what is the probability that (a) at least 10 survive,
(b) from 3 to 8 survive,
and (c) exactly 5 survive?

Solution: Let X be the number of people who survive.

$$\begin{aligned} \text{(a)} \quad P(X \geq 10) &= 1 - P(X < 10) = 1 - \sum_{x=0}^9 b(x; 15, 0.4) = 1 - 0.9662 \\ &= 0.0338 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(3 \leq X \leq 8) &= \sum_{x=3}^8 b(x; 15, 0.4) = \sum_{x=0}^8 b(x; 15, 0.4) - \sum_{x=0}^2 b(x; 15, 0.4) \\ &= 0.9050 - 0.0271 = 0.8779 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad P(X = 5) &= b(5; 15, 0.4) = \sum_{x=0}^5 b(x; 15, 0.4) - \sum_{x=0}^4 b(x; 15, 0.4) \\ &= 0.4032 - 0.2173 = 0.1859 \end{aligned}$$

Table A.1 (continued) Binomial Probability Sums $\sum_{x=0}^r b(x; n, p)$

<i>n</i>	<i>r</i>	<i>p</i>									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
15	0	0.2059	0.0352	0.0134	0.0047	0.0005	0.0000				
	1	0.5490	0.1671	0.0802	0.0353	0.0052	0.0005	0.0000			
	2	0.8159	0.3980	0.2361	0.1268	0.0271	0.0037	0.0003	0.0000		
	3	0.9444	0.6482	0.4613	0.2969	0.0905	0.0176	0.0019	0.0001		
	4	0.9873	0.8358	0.6865	0.5155	0.2173	0.0592	0.0093	0.0007	0.0000	
	5	0.9978	0.9389	0.8516	0.7216	0.4032	0.1509	0.0338	0.0037	0.0001	
	6	0.9997	0.9819	0.9434	0.8689	0.6098	0.3036	0.0950	0.0152	0.0008	
	7	1.0000	0.9958	0.9827	0.9500	0.7869	0.5000	0.2131	0.0500	0.0042	0.0000
	8		0.9992	0.9958	0.9848	0.9050	0.6964	0.3902	0.1311	0.0181	0.0003
	9		0.9999	0.9992	0.9963	0.9662	0.8491	0.5968	0.2784	0.0611	0.0022
	10		1.0000	0.9999	0.9993	0.9907	0.9408	0.7827	0.4845	0.1642	0.0127
	11			1.0000	0.9999	0.9981	0.9824	0.9095	0.7031	0.3518	0.0556
	12				1.0000	0.9997	0.9963	0.9729	0.8732	0.6020	0.1841
	13					1.0000	0.9995	0.9948	0.9647	0.8329	0.4510
	14						1.0000	0.9995	0.9953	0.9648	0.7941
	15							1.0000	1.0000	1.0000	1.0000



Table A.1 Binomial Probability Sums $\sum_{x=0}^r b(x; n, p)$

<i>n</i>	<i>r</i>	<i>p</i>									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
1	0	0.9000	0.8000	0.7500	0.7000	0.6000	0.5000	0.4000	0.3000	0.2000	0.1000
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0	0.8100	0.6400	0.5625	0.4900	0.3600	0.2500	0.1600	0.0900	0.0400	0.0100
	1	0.9900	0.9600	0.9375	0.9100	0.8400	0.7500	0.6400	0.5100	0.3600	0.1900
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.7290	0.5120	0.4219	0.3430	0.2160	0.1250	0.0640	0.0270	0.0080	0.0010
	1	0.9720	0.8960	0.8438	0.7840	0.6480	0.5000	0.3520	0.2160	0.1040	0.0280
	2	0.9990	0.9920	0.9844	0.9730	0.9360	0.8750	0.7840	0.6570	0.4880	0.2710
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.5905	0.3277	0.2373	0.1681	0.0778	0.0313	0.0102	0.0024	0.0003	0.0000
	1	0.9185	0.7373	0.6328	0.5282	0.3370	0.1875	0.0870	0.0308	0.0067	0.0005
	2	0.9914	0.9421	0.8965	0.8369	0.6826	0.5000	0.3174	0.1631	0.0579	0.0086
	3	0.9995	0.9933	0.9844	0.9692	0.9130	0.8125	0.6630	0.4718	0.2627	0.0815
	4	1.0000	0.9997	0.9990	0.9976	0.9898	0.9688	0.9222	0.8319	0.6723	0.4095
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000



Example 5.3: A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

- (a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
- (b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

Solution: (a) Denote by X the number of defective devices among the 20. Then X follows a $b(x; 20, 0.03)$ distribution. Hence,

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - b(0; 20, 0.03) \\ &= 1 - (0.03)^0(1 - 0.03)^{20-0} = 0.4562. \end{aligned}$$

- (b) In this case, each shipment can either contain at least one defective item or not. Hence, testing of each shipment can be viewed as a Bernoulli trial with $p = 0.4562$ from part (a). Assuming independence from shipment to shipment and denoting by Y the number of shipments containing at least one defective item, Y follows another binomial distribution $b(y; 10, 0.4562)$. Therefore,

$$P(Y = 3) = \binom{10}{3} 0.4562^3 (1 - 0.4562)^7 = 0.1602.$$

Since the probability distribution of any binomial random variable depends only on the values assumed by the parameters n , p , and q , it would seem reasonable to assume that the mean and variance of a binomial random variable also depend on the values assumed by these parameters.

Indeed, this is true, and in the proof of Theorem 5.1 we derive general formulas that can be used to compute the mean and variance of any binomial random variable as functions of n , p , and q .

Theorem 5.1:

The mean and variance of the binomial distribution $b(x; n, p)$ are
$$\mu = np \text{ and } \sigma^2 = npq.$$

Proof:

Let the outcome on the j th trial be represented by a Bernoulli random variable I_j , which assumes the values 0 and 1 with probabilities q and p , respectively.



Therefore, in a binomial experiment the number of successes can be written as the sum of the n independent indicator variables. Hence,

$$X = I_1 + I_2 + \cdots + I_n.$$

The mean of any I_j is $E(I_j) = (0)(q) + (1)(p) = p$.

Therefore, using Corollary 4.4 on page 131, the mean of the binomial distribution is

$$\mu = E(X) = E(I_1) + E(I_2) + \cdots + E(I_n) = \underbrace{p + p + \cdots + p}_{n \text{ terms}} = np.$$

The variance of any I_j is $\sigma_{I_j}^2 = E(I_j^2) - p^2 = (0)^2(q) + (1)^2(p) - p^2 = p(1 - p) = pq$.

Extending Corollary 4.11 to the case of n independent Bernoulli variables gives the variance of the binomial distribution as

$$\sigma_X^2 = \sigma_{I_1}^2 + \sigma_{I_2}^2 + \cdots + \sigma_{I_n}^2 = \underbrace{pq + pq + \cdots + pq}_{n \text{ terms}} = npq.$$



Example 5.4: It is conjectured that an impurity exists in 30% of all drinking wells in a certain rural community.

In order to gain some insight into the true extent of the problem, it is determined that some testing is necessary.

It is too expensive to test all of the wells in the area, so 10 are randomly selected for testing.

- (a) Using the binomial distribution, what is the probability that exactly 3 wells have the impurity, assuming that the conjecture is correct?
- (b) What is the probability that more than 3 wells are impure?

Solution: (a) We require

$$b(3; 10, 0.3) = \sum_{x=0}^3 b(x; 10, 0.3) - \sum_{x=0}^2 b(x; 10, 0.3) = 0.6496 - 0.3828 = 0.2668.$$

(b) In this case, $P(X > 3) = 1 - 0.6496 = 0.3504$.



There are solutions in which the computation of binomial probabilities may allow us to draw a scientific inference about population after data are collected. An illustration is given in the next example.



Example 5.6: Consider the situation of Example 5.4. The notion that 30% of the wells are impure is merely a conjecture put forth by the area water board.

Suppose 10 wells are randomly selected and 6 are found to contain the impurity.

What does this imply about the conjecture?

Use a probability statement.

Solution: We must first ask: “If the conjecture is correct, is it likely that we would find 6 or more impure wells?”

$$P(X \geq 6) = \sum_{x=6}^{10} b(x; 10, 0.3) = 1 - \sum_{x=0}^5 b(x; 10, 0.3) = 1 - 0.9527 = 0.0473.$$

As a result, it is very unlikely (4.7% chance) that 6 or more wells would be found impure if only 30% of all are impure.

This casts considerable doubt on the conjecture and suggests that the impurity problem is much more severe. 



As the reader should realize by now, in many applications there are more than two possible outcomes.

To borrow an example from the field of genetics, the color of guinea pigs produced as offspring may be red, black, or white.

Often the “defective” or “not defective” dichotomy is truly an oversimplification in engineering situations.

Indeed, there are often more than two categories that characterize items or parts coming off an assembly line.



Multinomial Experiments and the Multinomial Distribution

The binomial experiment becomes a **multinomial experiment** if we let each trial have more than two possible outcomes.

The drawing of a card from a deck *with replacement* is a multinomial experiment if the 4 suits are the outcomes of interest.

In general, if a given trial can result in any one of k possible outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the **multinomial distribution** will give the probability that E_1 occurs x_1 times, E_2 occurs x_2 times, $\dots$, and E_k occurs x_k times in n independent trials, where

$$x_1 + x_2 + \dots + x_k = n.$$

We shall denote this joint probability distribution by

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n).$$

Clearly, $p_1 + p_2 + \dots + p_k = 1$, since the result of each trial must be one of the k possible outcomes.



To derive the general formula, we proceed as in the binomial case.

Since the trials are independent, any specified order yielding x_1 outcomes for E_1 , x_2 for $E_2, \dots, x_k$ for E_k will occur with probability $p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$.

The total number of orders yielding similar outcomes for the n trials is equal to the number of partitions of n items into k groups with x_1 in the first group, x_2 in the second group, $\dots$, and x_k in the k th group.

This can be done in $\binom{n}{x_1, x_2, \dots, x_k} = \frac{n!}{x_1! x_2! \cdots x_k!}$ ways.

Since all the partitions are mutually exclusive and occur with equal probability, we obtain the multinomial distribution by multiplying the probability for a specified order by the total number of partitions.



Multinomial Distribution

If a given trial can result in the k outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of occurrences for $E_1, E_2, \dots, E_k$ in n independent trials, is

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

with

$$\sum_{i=1}^k x_i = n \text{ and } \sum_{i=1}^k p_i = 1.$$

The multinomial distribution derives its name from the fact that the terms of the multinomial expansion of $(p_1 + p_2 + \cdots + p_k)^n$ correspond to all the possible values of $f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n)$.



Example 5.7:

The complexity of arrivals and departures of planes at an airport is such that computer simulation is often used to model the “ideal” conditions.

For a certain airport with three runways, it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

$$\begin{aligned}\text{Runway 1: } p_1 &= 2/9, \\ \text{Runway 2: } p_2 &= 1/6, \\ \text{Runway 3: } p_3 &= 11/18.\end{aligned}$$

What is the probability that 6 randomly arriving airplanes are distributed in the following fashion?

$$\begin{aligned}\text{Runway 1: } & 2 \text{ airplanes,} \\ \text{Runway 2: } & 1 \text{ airplane,} \\ \text{Runway 3: } & 3 \text{ airplanes}\end{aligned}$$

Solution:

Using the multinomial distribution, we have

$$\begin{aligned}f\left(2, 1, 3; \frac{2}{9}, \frac{1}{6}, \frac{11}{18}, 6\right) &= \binom{6}{2, 1, 3} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3 \\ &= \frac{6!}{2! 1! 3!} \cdot \frac{2^2}{9^2} \cdot \frac{1}{6} \cdot \frac{11^3}{18^3} = 0.1127.\end{aligned}$$



Probability & Random Processes EE2011

Department of Electrical Engineering

Hypergeometric Distribution & Multivariate Hypergeometric Distribution

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



5.3 Hypergeometric Distribution

The types of applications for the hypergeometric are very similar to those for the binomial distribution.

in the case of the binomial distribution, independence among trials is required.

As a result, if that distribution is applied to, say, sampling from a lot of items (deck of cards, batch of production items), the sampling must be done **with replacement** of each item after it is observed.

On the other hand, the hypergeometric distribution does not require independence and is based on sampling done **without replacement**.



If we wish to find the probability of observing 3 red cards in 5 draws from an ordinary deck of 52 playing cards, the binomial distribution of Section 5.2 does not apply unless each card is replaced and the deck reshuffled before the next draw is made.

To solve the problem of sampling without replacement, let us restate the problem.

If 5 cards are drawn at random, we are interested in the probability of selecting 3 red cards from the 26 available in the deck and 2 black cards from the 26 available in the deck.



There are $\binom{26}{3}$ ways of selecting 3 red cards, and for each of these ways we can choose 2 black cards in $\binom{26}{2}$ ways.

Therefore, the total number of ways to select 3 red and 2 black cards in 5 draws is the product $\binom{26}{3}\binom{26}{2}$.

The total number of ways to select any 5 cards from the 52 that are available is $\binom{52}{5}$.

Hence, the probability of selecting 5 cards without replacement of which 3 are red and 2 are black is given by

$$\frac{\binom{26}{3}\binom{26}{2}}{\binom{52}{5}} = \frac{(26!/3!23!)(26!/2!24!)}{52!/5!47!} = 0.3251.$$



In general, we are interested in the probability of selecting x successes from the k items labeled successes and $n - x$ failures from the $N - k$ items labeled failures when a random sample of size n is selected from N items.

This is known as a **hypergeometric experiment**, that is, one that possesses the following two properties:

1. A random sample of size n is selected without replacement from N items.
2. Of the N items, k may be classified as successes and $N - k$ are classified as failures.

The number X of successes of a hypergeometric experiment is called a **hypergeometric random variable**.

Accordingly, the probability distribution of the hypergeometric variable is called the **hypergeometric distribution**, and its values are denoted by $h(x; N, n, k)$, since they depend on the number of successes k in the set N from which we select n items.



Let us now generalize in order to find a formula for $h(x; N, n, k)$.

The total number of samples of size n chosen from N items is $\binom{N}{n}$.

These samples are assumed to be equally likely.

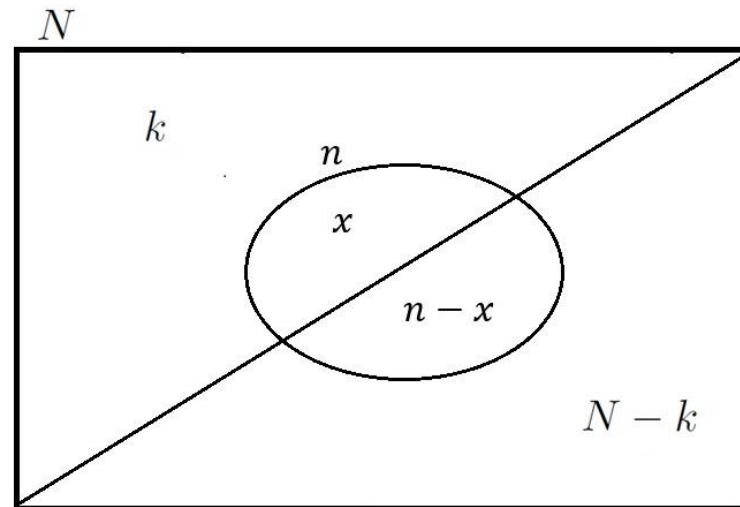
There are $\binom{k}{x}$ ways of selecting x successes from the k that are available, and for each of these ways we can choose the $n - x$ failures in $\binom{N-k}{n-x}$ ways.

Thus, the total number of favorable samples among the $\binom{N}{n}$ possible samples is given by $\binom{k}{x} \binom{N-k}{n-x}$.

Hence, we have the following definition.



Hypergeometric Distribution



random variable X , the number of successes in a random sample of size n selected from N items k are labeled **success** and $N - k$ labeled **failure**



Hypergeometric Distribution

The probability distribution of the hypergeometric random variable X , the number of successes in a random sample of size n selected from N items of which k are labeled **success** and $N - k$ labeled **failure**, is

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad \max\{0, n - (N - k)\} \leq x \leq \min\{n, k\}.$$

The range of x can be determined by the three binomial coefficients in the definition, where x and $n - x$ are no more than k and $N - k$, respectively, and both of them cannot be less than 0.

Usually, when both k (the number of successes) and $N - k$ (the number of failures) are larger than the sample size n , the range of a hypergeometric random variable will be $x = 0, 1, \dots, n$.



Theorem 5.2:

The mean and variance of the hypergeometric distribution $h(x; N, n, k)$ are

$$\mu = \frac{nk}{N} \text{ and } \sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right).$$

The proof for the mean is shown in Appendix A.24.



The hypergeometric distribution can be extended to treat the case where the N items can be partitioned into k cells $A_1, A_2, \dots, A_k$ with a_1 elements in the first cell, a_2 elements in the second cell, $\dots$, a_k elements in the k th cell.

We are now interested in the probability that a random sample of size n yields x_1 elements from A_1 , x_2 elements from A_2 , $\dots$, and x_k elements from A_k .

Let us represent this probability by

$$f(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n).$$

To obtain a general formula, we note that the total number of samples of size n that can be chosen from N items is still $\binom{N}{n}$.

There are $\binom{a_1}{x_1}$ ways of selecting x_1 items from the items in A_1 , and for each of these we can choose x_2 items from the items in A_2 in $\binom{a_2}{x_2}$ ways.



Therefore, we can select x_1 items from A_1 and x_2 items from A_2 in $\binom{a_1}{x_1} \binom{a_2}{x_2}$ ways.

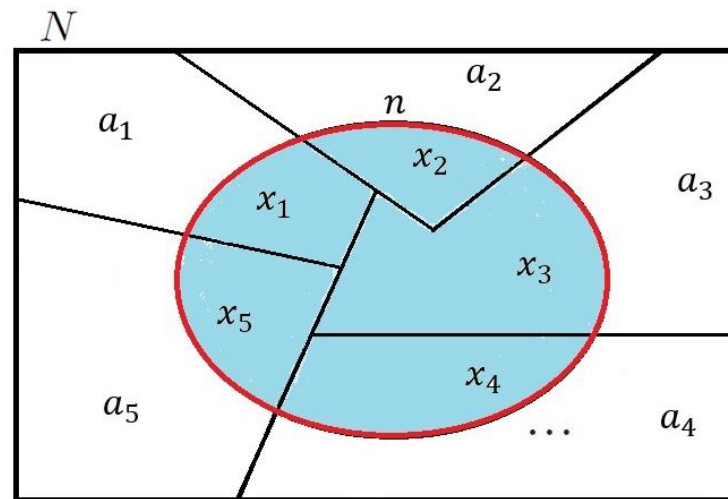
Continuing in this way, we can select all n items consisting of x_1 from A_1 , x_2 from A_2 , $\dots$, and x_k from A_k in

$$\binom{a_1}{x_1} \binom{a_2}{x_2} \dots \binom{a_k}{x_k} \text{ ways.}$$

The required probability distribution is now defined as follows.



Multivariate Hypergeometric Distribution



k cells $A_1, A_2, \dots, A_k$ with $a_1, a_2, \dots, a_k$ elements, respectively,
 $X_1, X_2, \dots, X_k$, representing the number of elements selected



Multivariate Hypergeometric Distribution

If N items can be partitioned into the k cells $A_1, A_2, \dots, A_k$ with $a_1, a_2, \dots, a_k$ elements, respectively, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of elements selected from $A_1, A_2, \dots, A_k$ in a random sample of size n , is

$$f(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n) = \frac{\binom{a_1}{x_1} \binom{a_2}{x_2} \dots \binom{a_k}{x_k}}{\binom{N}{n}},$$

with $\sum_{i=1}^k x_i = n$ and $\sum_{i=1}^k a_i = N$.



Example 5.8:

A particular part that is used as an injection device is sold in lots of 10.

The producer deems a lot acceptable if no more than one defective is in the lot.

A sampling plan involves random sampling and testing 3 of the parts out of 10.

If none of the 3 is defective, the lot is accepted.

Comment on the utility of this plan.

Solution:

Let us assume that the lot is truly **unacceptable** (i.e., that 2 out of 10 parts are defective).

The probability that the sampling plan finds the lot acceptable is

$$P(X = 0) = \frac{\binom{2}{0} \binom{8}{3}}{\binom{10}{3}} = 0.467.$$

Thus, if the lot is truly unacceptable, with 2 defective parts, this sampling plan will allow acceptance roughly 47% of the time. .

As a result, this plan should be considered faulty.



Example 5.9: Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives.

The procedure for sampling a lot is to select 5 components at random and to reject the lot if a defective is found.

What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?

Solution: Using the hypergeometric distribution with $n = 5$, $N = 40$, $k = 3$, and $x = 1$, we find the probability of obtaining 1 defective to be

$$h(1; 40, 5, 3) = \frac{\binom{3}{1} \binom{37}{4}}{\binom{40}{5}} = 0.3011.$$

Once again, this plan is not desirable since it detects a bad lot (3 defectives) only about 30% of the time.



Example 5.11:

Find the mean and variance of the random variable of Example 5.9 and the interval $\mu \pm 2\sigma$.

Solution:

Since Example 5.9 was a hypergeometric experiment with $N = 40$, $n = 5$, and $k = 3$, by Theorem 5.2, we have

$$\mu = \frac{nk}{N} \qquad \mu = \frac{(5)(3)}{40} = \frac{3}{8} = 0.375,$$

and

$$\sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right)$$
$$\sigma^2 = \left(\frac{40-5}{39}\right) (5) \left(\frac{3}{40}\right) \left(1 - \frac{3}{40}\right) = 0.3113.$$

Taking the square root of 0.3113, we find that $\sigma = 0.558$.

Hence, the required interval is $0.375 \pm (2)(0.558)$, or from -0.741 to 1.491 .



a binomial distribution can be used to approximate the hypergeometric distribution when n is small compared to N .

as a rule of thumb, the approximation is good when $n/N \leq 0.05$.

i.e., N is 20 times greater than n



Example 5.12:

A manufacturer of automobile tires reports that among a shipment of 5000 sent to a local distributor, 1000 are slightly blemished.

If one purchases 10 of these tires at random from the distributor, what is the probability that exactly 3 are blemished?

Solution:

Since $N = 5000$ is large relative to the sample size $n = 10$, we shall approximate the desired probability by using the binomial distribution.

The probability of obtaining a blemished tire is 0.2.

Therefore, the probability of obtaining exactly 3 blemished tires is

$$h(3; 5000, 10, 1000) \approx b(3; 10, 0.2) = 0.8791 - 0.6778 = 0.2013.$$

On the other hand, the exact probability is $h(3; 5000, 10, 1000) = 0.2015$. 



Example 5.13:

A group of 10 individuals is used for a biological case study.

The group contains 3 people with blood type O, 4 with blood type A, and 3 with blood type B.

What is the probability that a random sample of 5 will contain 1 person with blood type O, 2 people with blood type A, and 2 people with blood type B?

Solution:

Using the extension of the hypergeometric distribution with $x_1 = 1$, $x_2 = 2$, $x_3 = 2$, $a_1 = 3$, $a_2 = 4$, $a_3 = 3$, $N = 10$, and $n = 5$, we find that the desired probability is

$$f(1, 2, 2; 3, 4, 3, 10, 5) = \frac{\binom{3}{1} \binom{4}{2} \binom{3}{2}}{\binom{10}{5}} = \frac{3}{14}.$$



Probability & Random Processes EE2011

Department of Electrical Engineering

Negative Binomial and Geometric Distributions

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



Binomial Distribution

A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Theorem 5.1:

The mean and variance of the binomial distribution $b(x; n, p)$ are
 $\mu = np$ and $\sigma^2 = npq$.



Multinomial Distribution

If a given trial can result in the k outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of occurrences for $E_1, E_2, \dots, E_k$ in n independent trials, is

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

with

$$\sum_{i=1}^k x_i = n \text{ and } \sum_{i=1}^k p_i = 1.$$



Hypergeometric Distribution

The probability distribution of the hypergeometric random variable X , the number of successes in a random sample of size n selected from N items of which k are labeled **success** and $N - k$ labeled **failure**, is

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad \max\{0, n - (N - k)\} \leq x \leq \min\{n, k\}.$$

Theorem 5.2:

The mean and variance of the hypergeometric distribution $h(x; N, n, k)$ are

$$\mu = \frac{nk}{N} \text{ and } \sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right).$$

The proof for the mean is shown in Appendix A.24.



Multivariate Hypergeometric Distribution

If N items can be partitioned into the k cells $A_1, A_2, \dots, A_k$ with $a_1, a_2, \dots, a_k$ elements, respectively, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of elements selected from $A_1, A_2, \dots, A_k$ in a random sample of size n , is

$$f(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n) = \frac{\binom{a_1}{x_1} \binom{a_2}{x_2} \dots \binom{a_k}{x_k}}{\binom{N}{n}},$$

with $\sum_{i=1}^k x_i = n$ and $\sum_{i=1}^k a_i = N$.



5.4 Negative Binomial and Geometric Distributions

Let us consider an experiment where the properties are the same as those listed for a binomial experiment, with the exception that the trials will be repeated until a *fixed* number of successes occur.

Therefore, instead of the probability of x successes in n trials, where n is fixed, we are now interested in the probability that the k th success occurs on the x th trial.

Experiments of this kind are called **negative binomial experiments**.

As an illustration, consider the use of a drug that is known to be effective in 60% of the cases where it is used.

The drug will be considered a success if it is effective in bringing some degree of relief to the patient.

We are interested in finding the probability that the fifth patient to experience relief is the seventh patient to receive the drug during a given week.

Designating a success by S and a failure by F , a possible order of achieving the desired result is $SFSSSFS$, which occurs with probability

$$(0.6)(0.4)(0.6)(0.6)(0.6)(0.4)(0.6) = (0.6)^5(0.4)^2.$$



We could list all possible orders by rearranging the F 's and S 's except for the last outcome, which must be the fifth success.

The total number of possible orders is equal to the number of partitions of the first six trials into two groups with 2 failures assigned to the one group and 4 successes assigned to the other group.

$$\binom{6}{4} = 15 \text{ mutually exclusive ways.}$$

$$\begin{array}{l} SFSSSF \ S \\ FSFSSS \ S \\ SSFSFS \ S \end{array}$$

$$\binom{x-1}{k-1}$$

This can be done in $\binom{6}{4} = 15$ mutually exclusive ways.

Hence, if X represents the outcome on which the fifth success occurs,

then

$$P(X = 7) = \binom{6}{4} (0.6)^5 (0.4)^2 = 0.1866.$$



What Is the Negative Binomial Random Variable?

The number X of trials required to produce k successes in a negative binomial experiment is called a **negative binomial random variable**, and its probability distribution is called the **negative binomial distribution**.

Since its probabilities depend on the number of successes desired and the probability of a success on a given trial, we shall denote them by $b^*(x; k, p)$.

To obtain the general formula for $b^*(x; k, p)$, consider the probability of a success on the x th trial preceded by $k - 1$ successes and $x - k$ failures in some specified order.

Since the trials are independent, we can multiply all the probabilities corresponding to each desired outcome.

Each success occurs with probability p and each failure with probability $q = 1 - p$.

Therefore, the probability for the specified order ending in success is

$$p^{k-1} q^{x-k} p = p^k q^{x-k}.$$



The total number of sample points in the experiment ending in a success, after the occurrence of $k - 1$ successes and $x - k$ failures in any order, is equal to the number of partitions of $x - 1$ trials into two groups with $k - 1$ successes corresponding to one group and $x - k$ failures corresponding to the other group.

This number is specified by the term $\binom{x-1}{k-1}$, each mutually exclusive and occurring with equal probability $p^k q^{x-k}$.

We obtain the general formula by multiplying $p^k q^{x-k}$ by $\binom{x-1}{k-1}$.

**Negative
Binomial
Distribution**

If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable X , the number of the trial on which the k th success occurs, is

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x = k, k+1, k+2, \dots$$



The negative binomial distribution derives its name from the fact that each term in the expansion of $p^k(1 - q)^{-k}$ corresponds to the values of $b^*(x; k, p)$ for $x = k, k + 1, k + 2, \dots$

If we consider the special case of the negative binomial distribution where $k = 1$, we have a probability distribution for the number of trials required for a single success.

An example would be the tossing of a coin until a head occurs.

We might be interested in the probability that the first head occurs on the fourth toss.

The negative binomial distribution reduces to the form

$$b^*(x; 1, p) = pq^{x-1}, \quad x = 1, 2, 3, \dots$$



Since the successive terms constitute a geometric progression, it is customary to refer to this special case as the **geometric distribution** and denote its values by $g(x; p)$.

**Geometric
Distribution**

If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable X , the number of the trial on which the first success occurs, is

$$g(x; p) = pq^{x-1}, \quad x = 1, 2, 3, \dots$$

The following theorem states without proof the mean and variance of the geometric distribution.

Theorem 5.3:

The mean and variance of a random variable following the geometric distribution are

$$\mu = \frac{1}{p} \text{ and } \sigma^2 = \frac{1-p}{p^2}.$$



Example 5.15: For a certain manufacturing process, it is known that, on the average, 1 in every 100 items is defective.

What is the probability that the fifth item inspected is the first defective item found?

Solution: Using the geometric distribution with $x = 5$ and $p = 0.01$, we have

$$g(5; 0.01) = (0.01)(0.99)^4 = 0.0096.$$



Example 5.16: At a “busy time,” a telephone exchange is very near capacity, so callers have difficulty placing their calls.

It may be of interest to know the number of attempts necessary in order to make a connection.

Suppose that we let $p = 0.05$ be the probability of a connection during a busy time.

We are interested in knowing the probability that 5 attempts are necessary for a successful call.

Solution: Using the geometric distribution with $x = 5$ and $p = 0.05$ yields

$$P(X = x) = g(5; 0.05) = (0.05)(0.95)^4 = 0.041.$$



Applications of Negative Binomial and Geometric Distributions

In the case of the geometric distribution, Example 5.16 depicts a situation where engineers or managers are attempting to determine how inefficient a telephone exchange system is during busy times.

Clearly, in this case, trials occurring prior to a success represent a cost.

If there is a high probability of several attempts being required prior to making a connection, then plans should be made to redesign the system.

Applications of the negative binomial distribution are similar in nature. Suppose attempts are costly in some sense and are *occurring in sequence*.

A high probability of needing a “large” number of attempts to experience a fixed number of successes is not beneficial to the scientist or engineer.



Probability & Random Processes EE2011

Department of Electrical Engineering

Poisson Distribution

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



5.5 Poisson Distribution and the Poisson Process

Experiments yielding numerical values of a random variable X , the number of outcomes occurring during a given time interval or in a specified region, are called **Poisson experiments**.

The given time interval may be of any length, such as a minute, a day, a week, a month, or even a year.

For example, a Poisson experiment can generate observations for the random variable X representing the number of telephone calls received per hour by an office, the number of days school is closed due to snow during the winter, or the number of games postponed due to rain during a baseball season.

The specified region could be a line segment, an area, a volume, or perhaps a piece of material.

In such instances, X might represent the number of field mice per acre, the number of bacteria in a given culture, or the number of typing errors per page.

A Poisson experiment is derived from the **Poisson process** and possesses the following properties.



Properties of the Poisson Process

1. The number of outcomes occurring in one time interval or specified region of space is independent of the number that occur in any other disjoint time interval or region. In this sense we say that the Poisson process has no memory.
2. The probability that a single outcome will occur during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of outcomes occurring outside this time interval or region.
3. The probability that more than one outcome will occur in such a short time interval or fall in such a small region is negligible.

The number X of outcomes occurring during a Poisson experiment is called a **Poisson random variable**, and its probability distribution is called the **Poisson distribution**.

The mean number of outcomes is computed from $\mu = \lambda t$, where t is the specific “time,” “distance,” “area,” or “volume” of interest. Since the probabilities depend on λ , the rate of occurrence of outcomes, we shall denote them by $p(x; \lambda t)$.



The following formula is used for computing Poisson probabilities.

Poisson Distribution

The probability distribution of the Poisson random variable X , representing the number of outcomes occurring in a given time interval or specified region denoted by t , is

$$p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where λ is the average number of outcomes per unit time, distance, area, or volume and $e = 2.71828 \dots$.

Table A.2 contains Poisson probability sums,

$$P(r; \lambda t) = \sum_{x=0}^r p(x; \lambda t)$$

for selected values of λt ranging from 0.1 to 18.0.

We illustrate the use of this table with the following two examples.



Example 5.17: During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

Solution: Using the Poisson distribution with $x = 6$ and $\lambda t = 4$ and referring to Table A.2, we have

$$p(6; 4) = \frac{e^{-4}4^6}{6!} = \sum_{x=0}^6 p(x; 4) - \sum_{x=0}^5 p(x; 4) = 0.8893 - 0.7851 = 0.1042.$$

Table A.2 Poisson Probability Sums $\sum_{x=0}^r p(x; \mu)$

r	μ								
	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
0	0.3679	0.2231	0.1353	0.0821	0.0498	0.0302	0.0183	0.0111	0.0067
1	0.7358	0.5578	0.4060	0.2873	0.1991	0.1359	0.0916	0.0611	0.0404
2	0.9197	0.8088	0.6767	0.5438	0.4232	0.3208	0.2381	0.1736	0.1247
3	0.9810	0.9344	0.8571	0.7576	0.6472	0.5366	0.4335	0.3423	0.2650
4	0.9963	0.9814	0.9473	0.8912	0.8153	0.7254	0.6288	0.5321	0.4405
5	0.9994	0.9955	0.9834	0.9580	0.9161	0.8576	0.7851	0.7029	0.6160
6	0.9999	0.9991	0.9955	0.9858	0.9665	0.9347	0.8893	0.8311	0.7622
7	1.0000	0.9998	0.9989	0.9958	0.9881	0.9733	0.9489	0.9134	0.8666



Example 5.18: Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?

Solution: Let X be the number of tankers arriving each day. Then, using Table A.2, we have

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{x=0}^{15} p(x; 10) = 1 - 0.9513 = 0.0487.$$

Table A.2 (continued) Poisson Probability Sums $\sum_{x=0}^r p(x; \mu)$

r	μ								
	10.0	11.0	12.0	13.0	14.0	15.0	16.0	17.0	18.0
0	0.0000	0.0000	0.0000						
11	0.6968	0.5793	0.4616	0.3532	0.2600	0.1848	0.1270	0.0847	0.0549
12	0.7916	0.6887	0.5760	0.4631	0.3585	0.2676	0.1931	0.1350	0.0917
13	0.8645	0.7813	0.6815	0.5730	0.4644	0.3632	0.2745	0.2009	0.1426
14	0.9165	0.8540	0.7720	0.6751	0.5704	0.4657	0.3675	0.2808	0.2081
15	0.9513	0.9074	0.8444	0.7636	0.6694	0.5681	0.4667	0.3715	0.2867



Like the binomial distribution, the Poisson distribution is used for quality control, quality assurance, and acceptance sampling.

Theorem 5.4:

Both the mean and the variance of the Poisson distribution $p(x; \lambda t)$ are λt .



Nature of the Poisson Probability Function

Like so many discrete and continuous distributions, the form of the Poisson distribution becomes more and more symmetric, even bell-shaped, as the mean grows large.

Figure 5.1 illustrates this, showing plots of the probability function for $\mu = 0.1$, $\mu = 2$, and $\mu = 5$.

Note the nearness to symmetry when μ becomes as large as 5.

A similar condition exists for the binomial distribution, as will be illustrated later in the text.

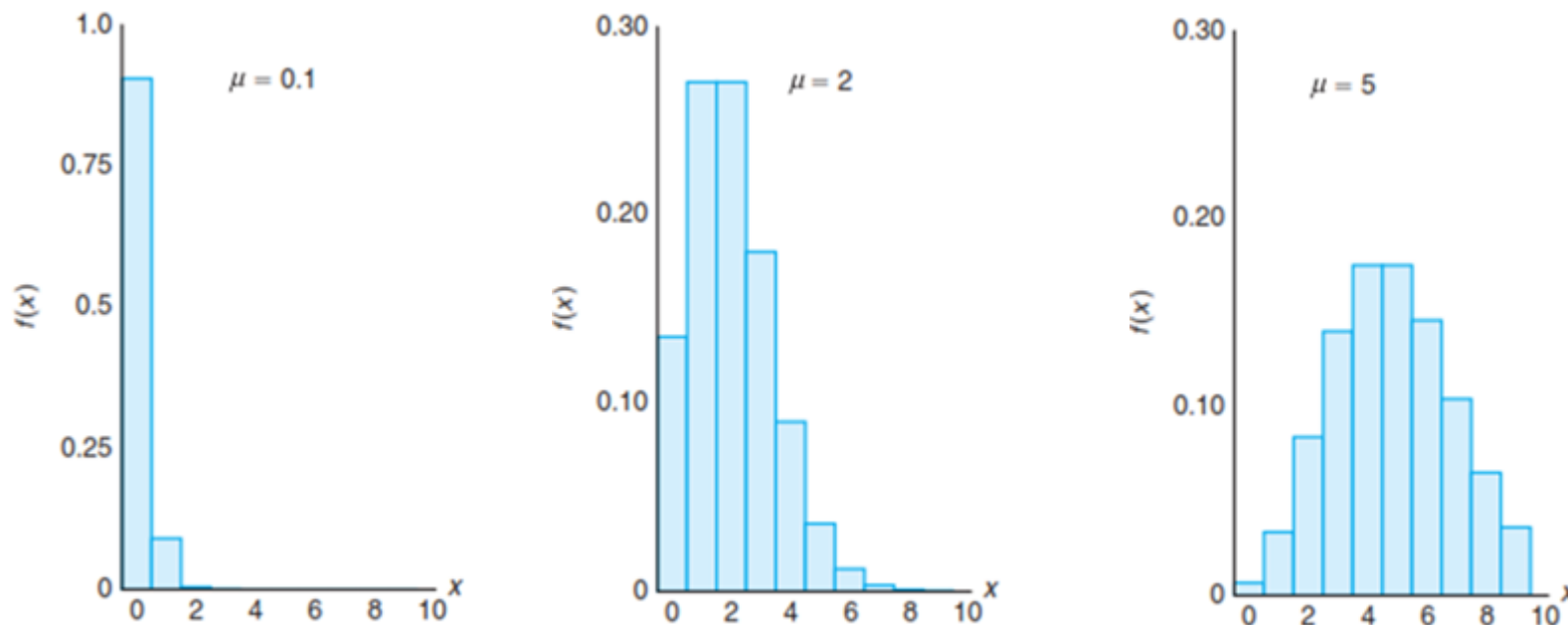


Figure 5.1: Poisson density functions for different means.



Approximation of Binomial Distribution by a Poisson Distribution

It should be evident from the three principles of the Poisson process that the Poisson distribution is related to the binomial distribution.

Although the Poisson usually finds applications in space and time problems, as illustrated by Examples 5.17 and 5.18, it can be viewed as a limiting form of the binomial distribution.

In the case of the binomial, if n is quite large and p is small, the conditions begin to simulate the *continuous space or time* implications of the Poisson process.



Indeed, if n is large and p is close to 0, the Poisson distribution can be used, with $\mu = np$, to approximate binomial probabilities.

Theorem 5.5:

Let X be a binomial random variable with probability distribution $b(x; n, p)$. When $n \rightarrow \infty$, $p \rightarrow 0$, and $np \xrightarrow{n \rightarrow \infty} \mu$ remains constant,

$$b(x; n, p) \xrightarrow{n \rightarrow \infty} p(x; \mu).$$



Example 5.19: In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent of each other.

- (a) What is the probability that in any given period of 400 days there will be an accident on one day?
- (b) What is the probability that there are at most three days with an accident?

Solution: Let X be a binomial random variable with $n = 400$ and $p = 0.005$. Thus, $np = 2$.

Using the Poisson approximation,

(a) $P(X = 1) = e^{-2}2^1 = 0.271$ and

(b) $P(X \leq 3) = \sum_{x=0}^3 e^{-2}2^x/x! = 0.857.$



Example 5.20: In a manufacturing process where glass products are made, defects or bubbles occur, occasionally rendering the piece undesirable for marketing.

It is known that, on average, 1 in every 1000 of these items produced has one or more bubbles.

What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

Solution: This is essentially a binomial experiment with $n = 8000$ and $p = 0.001$.

Since p is very close to 0 and n is quite large, we shall approximate with the Poisson distribution using

$$\mu = (8000)(0.001) = 8.$$

Hence, if X represents the number of bubbles, we have

$$P(X < 7) = \sum_{x=0}^6 b(x; 8000, 0.001) \approx p(x; 8) = 0.3134.$$



3. A small business receives, on average, 8 calls per hour. (a) What is the probability that the business will receive exactly 7 calls in 1 hour? (b) What is the probability that the business will receive, at most, 5 calls in one hour? (c) What is the probability that the business will receive more than 6 calls in one hour?

$$\begin{aligned}\mu &= 8 \quad P(X > 6) \\ P(X > 6) &= 1 - P(X \leq 6) \\ &= 1 - [P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6)] \\ &= 1 - e^{-8} \left[\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} + \frac{8^6}{6!} \right]\end{aligned}$$

